

Micah Olivas

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Education

Ph.D. , Genetics Stanford University , Stanford, CA Advisor: Polly Fordyce Thesis: <i>High-throughput enzymology for protein sequence–function mapping</i>	2020–2026 (<i>expected</i>)
B.S. with Honors (<i>summa cum laude</i>), Biochemistry California State University, Fresno , Fresno, CA Advisor: Laurent Dejean Thesis: <i>Particulate matter-induced oxidative stress in alveolar macrophages</i>	2016–2020
Visiting Student , Chemistry and Cell Biology University of Oxford , Oxford, UK Host Advisor: Aziz Aboobaker	2019

Grants

F31HG013267 , National Human Genome Research Institute <i>High-throughput thermodynamic and kinetic measurements for variant effects prediction in a major protein superfamily</i> Role: PI; Funding: \$143,908	09/2023–08/2026
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Awards and honors

Selected graduate speaker (only graduate student), <i>Gordon Research Conference on Enzymes</i>	2025
American Chemical Society Poster Award, <i>Gordon Research Conference on Enzymes</i>	2024
Ruth L. Kirschstein National Research Service Award, <i>National Institutes of Health</i>	2023
Stanford ADVANCE Scholar	2020–2021
Graduate Fellowship, <i>Phi Kappa Phi</i>	2020
Marshall Scholarship Finalist	2020
Amgen Scholar, <i>Stanford University</i>	2019
Barry M. Goldwater Scholar	2019
Chemistry Department Honors, <i>Fresno State</i>	2019
Presidents Scholar, <i>Fresno State</i>	2016
Smittcamp Family Honors Scholar (full-tuition undergraduate scholarship)	2016

Publications

[†]denotes equal contributions

Manuscripts in preparation

1. P. Almhjell[†], M. Olivas[†], F. Zepezauer, D. Herschlag, and P. Fordyce. "Expansive and systematic perturbation of enzyme function via microfluidic variant assays with suppressor tRNAs." Manuscript in preparation.

2. M. Olivas[†], P. Almhjell[†], F. Zepezauer, D. Herschlag, and P. Fordyce. "uSort-M: scalable isolation of user-defined clones from pooled gene libraries." *ACS Synthetic Biology*, under review (2026). bioRxiv: [q3bk](#).

Peer-reviewed publications

1. K. Han, S. E. Pierce, A. Li, K. Spees, G. R. Anderson, J. A. Seoane, Y. H. Lo, M. Dubreuil, M. Olivas, *et al.* "CRISPR screens in cancer spheroids identify 3D growth-specific vulnerabilities." *Nature* 580 (2020), 136–141. DOI: [gg2bv3](#)

Conference papers

1. C. Fannjiang[†], M. Olivas[†], *et al.* "Designing active and thermostable enzymes with sequence-only predictive models." *NeurIPS Learning Meaningful Representations of Life workshop* (2022). OpenReview: [Nc7EspZ7C](#)

Patents

Pending

P. Almhjell, M. Olivas, & P. Fordyce. High-throughput production of protein variants.
U.S. Provisional Patent Application No. 63/916,150.

Contributed talks

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| Jul 2025 | Using high-throughput microfluidic enzymology to investigate machine learning for function prediction
<i>Gordon Research Conference on Enzymes. Waterville Valley, NH</i> |
| Apr 2025 | Reimagining mutational scanning through microfluidic variant assays with suppressor tRNAs
<i>Chan-Zuckerberg Biohub Interlab Meeting. San Francisco, CA</i> |
| Feb 2025 | Reimagining mutational scanning through variant assays with suppressor tRNAs
<i>Current Issues in Genetics. Stanford, CA</i> |
| May 2024 | Insights from high-throughput enzymology in a small enzyme
<i>Current Issues in Genetics. Stanford, CA</i> |
| Aug 2023 | Beyond Structure: Exploring the Language of Enzyme Function with HT-MEK
<i>Current Issues in Genetics. Stanford, CA</i> |
| Mar 2018 | Sewn in Your Genes: Reimagining Disease in the Age of Genetic Editing
<i>18th annual Smittcamp Family Honors College Colloquium. Fresno, CA</i> |

Contributed posters

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| Sep 2025 | Reimagining mutational scanning through microfluidic variant assays with suppressor tRNAs
<i>Genetics Department Retreat. Stanford, CA</i> |
| Jul 2025 | Using high-throughput microfluidic enzymology to investigate machine learning for function prediction
<i>Gordon Research Conference (GRC) on Enzymes. Waterville Valley, NH</i> |
| Jun 2025 | Reimagining mutational scanning through microfluidic variant assays with suppressor tRNAs
<i>Mutational Scanning Symposium. Barcelona, Spain</i> |
| Apr 2025 | Reimagining mutational scanning through microfluidic variant assays with suppressor tRNAs
<i>NHGRI Trainee Meeting. Philadelphia, PA</i> |
| Jul 2024 | Using high-throughput microfluidic enzymology to investigate machine learning for functional prediction
<i>Gordon Research Conference (GRC) on Enzymes. Waterville Valley, NH</i> |
| Sep 2023 | Leveraging novel protein language models to understand constraints on enzyme function and design
<i>Genetics Department Annual Retreat. Monterey, CA</i> |
| Nov 2022 | Leveraging novel protein language models to understand constraints on enzyme function and design
<i>Chan Zuckerberg Biohub Interlab Confabulation. San Francisco, CA</i> |

- Sep 2022 Leveraging novel protein language models to understand constraints on enzyme function and design
Quantitative Biology Institute Bay Area Chemical Biology Symposium. San Francisco, CA
- Sep 2021 Developing predictive models of enzyme function from physics and phylogeny
Genetics Department Annual Retreat. Stanford, CA
- Aug 2019 Genome-wide CRISPR screens in 3D tumor spheroids reveal growth dependencies in lung adenocarcinoma
Stanford Summer Research Program Symposium. Stanford, CA
- July 2018 Development of an inducible Cas9 for temporally controlled gene editing
BioCoRE symposium. Durham, NC
- Apr 2018 PM2.5 Exposure and ROS Production in NR8383 Alveolar Macrophages
CSU Honors Research Conference. Northridge, CA
- Jan 2018 PM2.5 Exposure and ROS Production in NR8383 Alveolar Macrophages
CSU Program for Education and Research in Biotechnology (CSUPERB). Santa Clara, CA
- Apr 2017 Effects of particulate matter aerosols on ROS production in alveolar macrophage cells
Central California Research Symposium. Fresno, CA

Academic service

Scientific mentorship

Winter 2025	Diego Pomales-Matos	Stanford Genetics Graduate Student
Summer 2024	Forrest Zepezauer	Summer Undergraduate Student